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Certified Technical Architect (CTA)

Program description and syllabus

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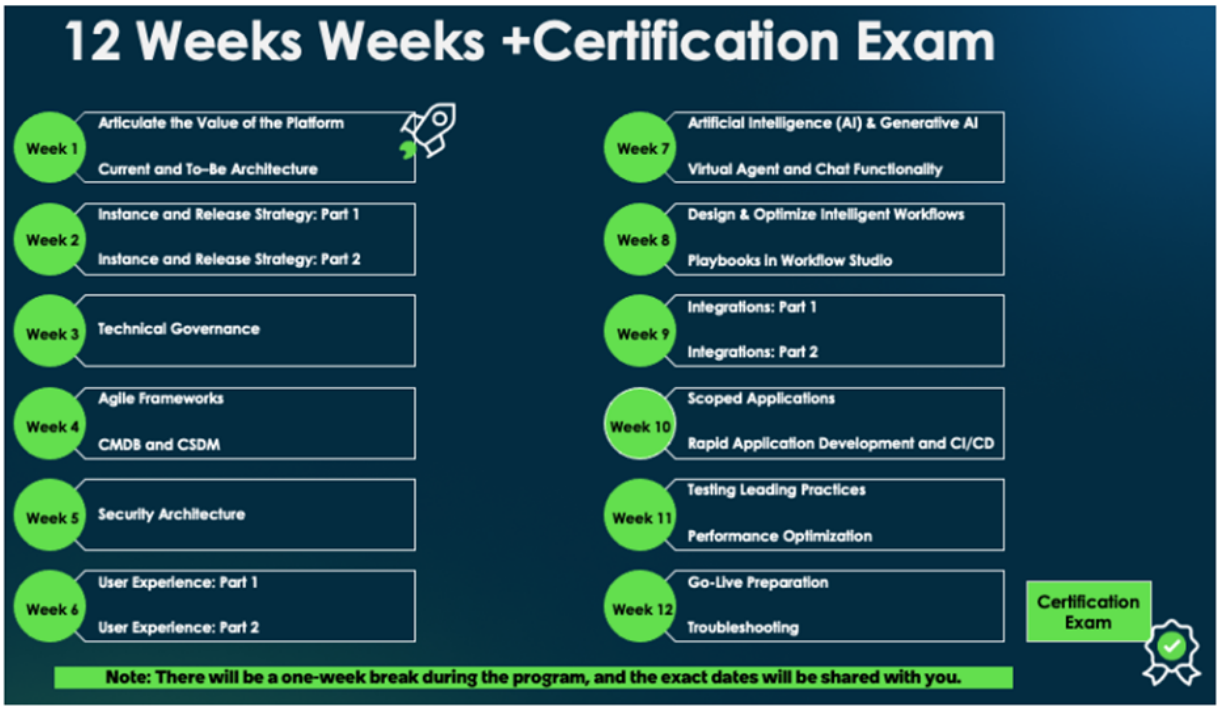
Overview

The Certified Technical Architect (CTA) Program develops your ability to solve complex business challenges using the ServiceNow platform. You will master integrating and orchestrating multiple systems while building expertise in technical governance processes for designing robust platform solutions. The program also strengthens your communication skills, preparing you to effectively present architectural solutions to stakeholders at all management levels and design elegant solutions that improve stakeholder experience.

Course format

Each week, you'll work through self-paced modules to build knowledge, then put it into practice through hands-on activities. On-demand content for the upcoming week will be made available every Friday.

A typical week includes a combination of self-study, individual and group activities, discussions, peer reviews, and virtual sessions. Learners may structure their self-study time according to personal and professional commitments. Groups are responsible for coordinating and agreeing on meeting times for collaborative study and completion of group assignments.



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Week 1

Module: Articulate the Value of the Platform

Objectives:

- Describe the importance of articulating the value of ServiceNow platform capabilities
- Identify benefits and supporting evidence for solutions and their capabilities
- Present the value of the platform and how it meets the customer's desired outcomes
- Tailor messaging about the platform's benefits based on the audience, whether addressing developers, business stakeholders, or operations teams

Activities:

Group activity: Analyze the GlobalPort Logistics scenario to design a value proposition that clearly articulates ServiceNow platform value, aligns with business needs, and resonates with internal stakeholders.

Individual activity (optional): Record your group presentation, communicating the solution to stakeholders.

Module: Current and To-Be Architecture

Objectives:

- List the four main domains of technical architecture implementations
- Apply the three steps to identify current architecture and to-be architecture
- Prepare for customer interviews to gather information, document findings, and create structured diagrams
- Organize information to present findings back to stakeholders

Activity:

Group activity: Conduct architecture assessments and design scalable to be architecture by evaluating ServiceNow capability maturity, identifying gaps, and aligning technical solutions with business priorities.

Week 2

Module: Instance and Release Strategy: Part 1

Objectives:

- Explain the key elements of governance frameworks that support instance and release strategies
- Describe the essential considerations for developing an effective instance policy
- Identify the benefits and challenges of multi-production, multi-development, and domain-separated instances

Module: Instance and Release Strategy: Part 2

Objectives:

- Explain the key benefits of Instance Data Replication (IDR) in ensuring data consistency across multiple instances
- Evaluate appropriate use cases for implementing IDR based on business and technical requirements
- Compare the benefits and challenges of different code migration methods when developing an effective ServiceNow release strategy

Activities:

Group activity: Analyze Acme Corporation's key challenges, such as latency, compliance, regulatory risks, maintenance, and resource constraints, and suggest whether a single global instance or multiple regional instances is the best fit.

Individual activity: Record your individual presentation, communicating the solution to stakeholders.

Week 3

Module: Technical Governance

Objectives:

- Explain the benefits of investing in ServiceNow governance and identify strategies for establishing effective governance

- Recall how the governance levels of technical, portfolio, and strategy are interrelated and should inform each other
- Guide customers in defining technical governance policies and standards that cover data management, environment setup, application development, and platform management in ServiceNow
- Identify key elements that should be prioritized to drive continuous improvement of the platform and the governance process

Activities:

Group activity: Design a technical governance board for Tech Forward Inc. to define, implement, and enforce technical standards and best practices to ensure effective decision-making and oversight.

Individual activity: Record your individual presentation, communicating the solution to stakeholders.

Week 4

Module: Agile Frameworks

Objectives:

- Identify different agile ways of working and apply them to implement and provide governance to a particular task or a project
- Translate factors that can help architects determine the leading methodology, or combination of methodologies, to apply to their customers

Module: CMDB and CSDM

Objectives:

- Apply the CMDB/CSDM data modelling process to help customers achieve their desired outcomes
- Explain key CSDM concepts, including applications, services, and capabilities
- Identify the three pillars of CMDB management and the core tools and processes that support each one

Activity:

Group activity: Analyze the Zenovia Life Sciences scenario to present recommendations on applying Agile frameworks to the project team and increasing their CMDB's alignment to the CSDM.

Week 5

Module: Security Architecture

Objectives:

- Describe the responsibilities of each party in the shared security model
- Identify the three layers of platform security
- Describe the security options available to organizations within each layer
- Determine when to use encryption to protect and secure data at rest

Activity:

Group activity (discussion board): Evaluate ServiceNow's proposed security architecture for GlobalPort Logistics. Your objective is to determine whether the recommended approach effectively addresses leadership priorities around confidentiality, integrity, availability, and compliance.

Week 6

Module: User Experience: Part 1

Objectives:

- Recall key principles of user experience design, such as empathy, validation, and reuse
- Apply common tools used early, during, and after designing and validating user experiences
- Identify content creation, user interface design, and interaction design leading practices for a user-friendly experience

Module: User Experience: Part 2

Objectives:

- Identify the components, technologies, and frameworks of the Service Portal

- Explain how widgets contribute to building dynamic user experiences
- Compare the tools used to determine Service Portal usage and health
- Apply content architecture and strategy principles within ServiceNow portals

Activity:

Group activity: Analyze the GlobalFin Services scenario to design UX mockups demonstrating an understanding of users' needs and analysis of pain points to propose areas for improvement.

Week 7

Module: Artificial Intelligence & Generative Artificial Intelligence

Objectives:

- Explain the differences between Artificial Intelligence and Generative Artificial Intelligence
- Describe ServiceNow's AI and GenAI offerings
- Map ServiceNow's Generative AI capabilities to ServiceNow's Now Assist products
- Describe the steps for installing and configuring Now Assist products
- Explain the workings of Now Assist frameworks and their underlying technologies
- Describe the differences between ServiceNow's Now Large Language Model (LLM) and third-party external LLMs

Activity:

Group activity: Analyze the GlobalBank Investments scenario to design Artificial Intelligence (AI)-powered automation using Virtual Agent, Predictive Intelligence, generative AI, and agentic AI.

Week 8

Module: Design and Optimize Intelligent Workflows

Objectives:

- Identify tools for designing and optimizing intelligent workflows
- Leverage available solutions to help operate intelligent workflows
- Select the best solutions to analyze intelligent workflows

Activity:

Group activity: Analyze Value Care Solutions' current workflow, highlight risks, and recommend strategies that optimize the workflows and align with business goals and KPI.

Module: Playbooks in Workflow Studio

Objectives:

- Recall Playbook key components and architecture
- Identify the runtime components produced by Playbooks
- Describe the six major steps involved in the process definition runtime
- Explain the benefits and limitations of Playbooks

Activity:

Group activity (discussion board): Evaluate whether Playbooks enhance clarity, accountability, and governance, and advise leadership on risks, gaps, and improvement opportunities.

Week 9

Module: Integrations: Part 1

Objectives:

- Describe the concept of IT system integrations
- Explain the benefits of integrations
- Identify options and considerations that impact integrations
- Identify leading practices for ServiceNow integrations

Module: Integrations: Part 2

Objectives:

- Identify ServiceNow foundation data principles and usage
- Explain how Service Portfolio Management is used to manage services data
- Apply the key factors in managing governance of foundation data
- Recognize integration patterns for choosing the correct integration
- Identify platform integration capabilities with external systems

Activities:

Group activity: Analyze TerraFusion's scenario to recommended integration approach, the methodology, and the rationale behind the decision, tailored to different stakeholders: Enterprise Architect, Platform Owner, and Data Owner.

Individual activity: Record your individual presentation, communicating the solution to stakeholders.

Week 10

Module: Scoped Applications

Objectives:

- Recall the process for developing and sharing scoped apps
- Identify aspects of cross scope privileges
- Review scoped application considerations prior to development

Module: Rapid Application Development & CI/CD

Objectives:

- Define Rapid Application Development (RAD), Continuous Integration/Continuous Delivery (CI/CD), and their foundational principles
- Explain strategies for minimizing automation sprawl
- Describe the benefits of RAD and CI/CD in developing and delivering applications quickly
- Identify the unique governance and compliance challenges in rapid development

Activity:

Group activity: Analyze the Novatech scenario to design a value proposition that clearly articulates ServiceNow platform value, aligns with business needs, and resonates with internal stakeholders.

Week 11

Module: Testing Leading Practices

Objectives:

- Define the main ServiceNow testing methods
- Identify key differences between manual, automated, or special test types

- Choose the best testing option based on leading practices

Module: Performance Optimization

Objectives:

- Describe how performance issues could arise and the importance of prevention
- Identify implementation areas that impact performance
- Explore methods and frequency for monitoring instance hygiene

Activities:

Group activity: Analyze the NextGen Dynamics scenario to identify gaps in the current testing strategy and develop a performance optimization strategy.

Individual activity: Record your individual presentation, communicating the solution to stakeholders.

Week 12

Module: Go-Live Preparation

Objectives:

- Describe the process and leading practices required before go-live
- Define the steps to manage organizational change, handle issues, and manage risks
- Plan for communication, address customer challenges, and offer post go-live support

Module: Troubleshooting

Objectives:

- Apply effective troubleshooting methods to common issues
- Identify leading practices for ongoing platform support
- Connect with ServiceNow support channels and community
- Recognize frequent MID server problems and their typical symptoms

Activities:

Group activity: Analyze the CloudVision Solutions scenario to evaluate their go-live strategy and build a MID Server troubleshooting guide.

Individual activity: Record your individual presentation, communicating the solution to stakeholders.